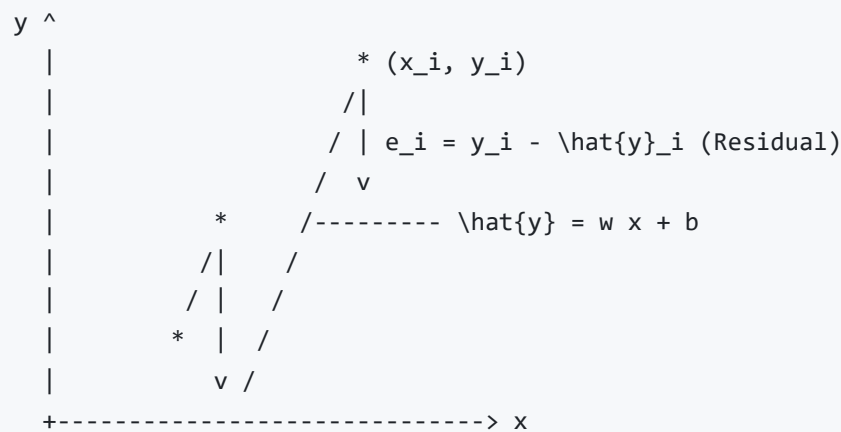


Module 23.08: What is the Best Fit Line? (Residuals & Ordinary Least Squares)

1. Introduction: The Criterion for "Best Fit"

Given a set of two-dimensional observation points $(x^{(1)}, y^{(1)}), (x^{(2)}, y^{(2)}), \dots, (x^{(m)}, y^{(m)})$, infinitely many candidate straight lines $\hat{y} = wx + b$ can be drawn through the scatter plot. Determining the **best fit line** requires establishing an objective mathematical criterion that measures how closely the candidate line approximates the empirical observations.



2. Defining the Residual (Error)

For any observation $(x^{(i)}, y^{(i)})$, the candidate line predicts:

$$\hat{y}^{(i)} = wx^{(i)} + b$$

The deviation between the empirical ground truth $y^{(i)}$ and the candidate prediction $\hat{y}^{(i)}$ is known as the **residual** (or error) $e^{(i)}$:

$$e^{(i)} = y^{(i)} - \hat{y}^{(i)} = y^{(i)} - (wx^{(i)} + b)$$

Why Not Minimize the Raw Sum of Residuals?

If we defined total error as $\sum_{i=1}^m e^{(i)}$:

- Positive residuals (points lying above the line) cancel out negative residuals (points lying below the line).
- A completely erratic line passing nowhere near the data points could have $\sum_{i=1}^m e^{(i)} = 0$ purely through destructive interference of positive and negative errors.

Why Not Minimize the Sum of Absolute Residuals (MAE / L1 Norm)?

If we minimized $\sum_{i=1}^m |e^{(i)}|$:

- The absolute value function $f(u) = |u|$ is non-differentiable at $u = 0$.
- It lacks a clean closed-form analytical derivative, making matrix inversion and standard calculus optimization mathematically cumbersome.

3. Ordinary Least Squares (OLS) Criterion

The foundational solution, established independently by Carl Friedrich Gauss and Adrien-Marie Legendre, is the **Ordinary Least Squares (OLS)** principle. We define the objective as minimizing the **Sum of Squared Errors (SSE)**:

$$SSE(w, b) = \sum_{i=1}^m \left(e^{(i)}\right)^2 = \sum_{i=1}^m \left(y^{(i)} - (wx^{(i)} + b)\right)^2$$

Mathematical Properties of OLS:

1. **Positivity:** Squaring guarantees $(e^{(i)})^2 \geq 0$. Errors cannot cancel out.
 2. **Disproportionate Penalty on Outliers:** A deviation of 4 units contributes $4^2 = 16$ to the error, whereas a deviation of 1 unit contributes $1^2 = 1$. OLS heavily penalizes large errors.
 3. **Smoothness and Differentiability:** The function $SSE(w, b)$ is continuously differentiable across all $w, b \in \mathbb{R}$, yielding a strictly convex paraboloid with an exact analytical minimum.
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4. Analytical Derivation of the OLS Closed-Form Solution

To find w and b that minimize $SSE(w, b)$, we set the partial derivatives with respect to b and w to zero.

Step 1: Derivative with Respect to Intercept b

$$\frac{\partial SSE}{\partial b} = \frac{\partial}{\partial b} \sum_{i=1}^m (y^{(i)} - wx^{(i)} - b)^2 = -2 \sum_{i=1}^m (y^{(i)} - wx^{(i)} - b) = 0$$

Dividing by $-2m$:

$$\frac{1}{m} \sum_{i=1}^m y^{(i)} - w \left(\frac{1}{m} \sum_{i=1}^m x^{(i)} \right) - b = 0 \implies \bar{y} - w\bar{x} - b = 0$$

Thus, the optimal intercept is:

$$b = \bar{y} - w\bar{x}$$

Crucial Geometric Insight: The optimal best fit line always passes through the sample mean centroid $(\bar{x}, \bar{y})$.

Step 2: Derivative with Respect to Slope w

Substitute $b = \bar{y} - w\bar{x}$ into the error term:

$$y^{(i)} - \hat{y}^{(i)} = y^{(i)} - (wx^{(i)} + \bar{y} - w\bar{x}) = (y^{(i)} - \bar{y}) - w(x^{(i)} - \bar{x})$$

Differentiating SSE with respect to w :

$$\frac{\partial SSE}{\partial w} = -2 \sum_{i=1}^m (x^{(i)} - \bar{x}) \left[(y^{(i)} - \bar{y}) - w(x^{(i)} - \bar{x}) \right] = 0$$

Distributing the summation:

$$\sum_{i=1}^m (x^{(i)} - \bar{x})(y^{(i)} - \bar{y}) - w \sum_{i=1}^m (x^{(i)} - \bar{x})^2 = 0$$

Solving for w :

$$w = \frac{\sum_{i=1}^m (x^{(i)} - \bar{x})(y^{(i)} - \bar{y})}{\sum_{i=1}^m (x^{(i)} - \bar{x})^2} = \frac{\text{Cov}(x, y)}{\text{Var}(x)}$$

5. Summary Table

Property	Symbol	Formula	Statistical Meaning
Residual	$e^{(i)}$	$y^{(i)} - \hat{y}^{(i)}$	Vertical error between truth and fit
Objective	SSE	$\sum_{i=1}^m (e^{(i)})^2$	Sum of squared vertical distances
Optimal Slope	w^*	$\frac{\text{Cov}(x, y)}{\text{Var}(x)}$	Ratio of sample covariance to sample feature variance
Optimal Intercept	b^*	$\bar{y} - w^* \bar{x}$	Ensures line passes through $(\bar{x}, \bar{y})$